

The Theater Machine: An Interactive Generative-AI-Based Installation for Participatory Cultural Mediation

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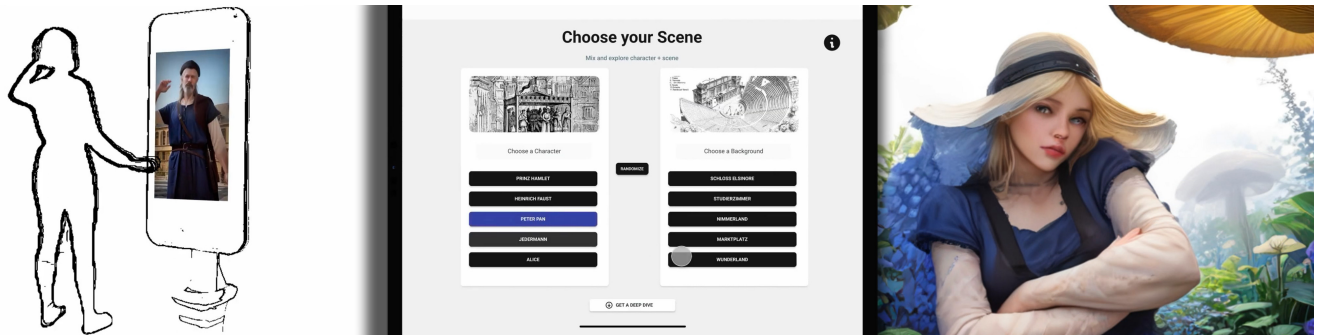


Figure 1: The concept of *The Theater-Machine*. A user selects a play, stage design, and character, and can control the chosen character via video feed. Left – installation sketch, center – user interface, right – video-feed output.

Abstract

The cooperative project *The Theater Machine* develops an interactive, AI-based installation for the foyer of a theater. It aims to lower barriers to performing arts by enabling participatory engagement with theatrical content. Users interact via camera and touchscreen, being integrated in real time into dynamically generated theater scenes. The system operates locally, using a high-performance workstation running a generative model based on Adversarial Diffusion Distillation within a high-throughput StreamDiffusion pipeline. This paper outlines the conceptual framework, system architecture, interaction design, and sustainability model.

CCS Concepts

• **Applied computing** → **Media arts**; • **Human-centered computing** → **Interaction techniques**; • **Computing methodologies** → **Computer vision**.

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1 Introduction and Conceptual Framework

The project *The Theater Machine* is an installation that enables users to place themselves, via real-time AI using a StreamDiffusion pipeline [Kodaira et al. 2023], into various theatrical productions and historical periods and is situated at the intersection of performing arts, digital cultural mediation, and human-computer interaction. The user positions themselves in front of a camera and, via a touchscreen interface, independently selects both a character and a setting from a curated set of options. The real-time AI system then transforms the user's appearance into the chosen character and places them within the selected scene. By integrating motion tracking, the system produces a real-time, AI-generated mirror image.

The project addresses the challenge of making traditional cultural forms accessible to an increasingly digitally oriented, diverse, and younger audience. Conventional theater reception is typically defined by a passive spectator role [Forenza 2017; Reason et al. 2015].

By enabling users to create their own theatrical scenes, the installation promotes deeper cognitive and emotional engagement with the material, thereby fostering more enduring cultural experiences [Wang et al. 2022].

A key element of the project is its collaborative, interdisciplinary model linking theater with playful media. The Landestheater Linz (State Theatre of Linz), Austria, contributes expertise in dramaturgy, stage design, and production infrastructure, while the University of Applied Sciences Upper Austria develops the AI-driven real-time image synthesis system. The central interaction concept is jointly developed to ensure close integration between artistic curation and technical feasibility. This structured division of labor is considered essential to the realization of the project [Shalley and Gilson 2017a].

2 Software Architecture and Data Flow

The software architecture is designed as a closed, local system. The internal data flow operates in a continuous loop. In contrast to other image-to-video models, the *Theater Machine* is capable of producing multiple frames per second, making it more reactive and engaging for the user [Shalley and Gilson 2017b].

The camera records a continuous video stream of the user in front of the station. Deep-learning algorithms use the Nvidia background remover to segment the user's image from the background. The touchscreen interface receives the user's selection (e.g., "Swan Lake"), which is converted into a text prompt and in some cases into an example picture for the AI model. The segmented user image and textual prompt are sent to the StreamDiffusion pipeline. The model replaces the background with an AI-generated scene matching the prompt and image and integrates the user image in a consistent visual style. The synthesized video frame is displayed on screen. This cycle repeats approximately 20 times per second, producing fluid, interactive video output at speeds exceeding standard *img2img* methods.

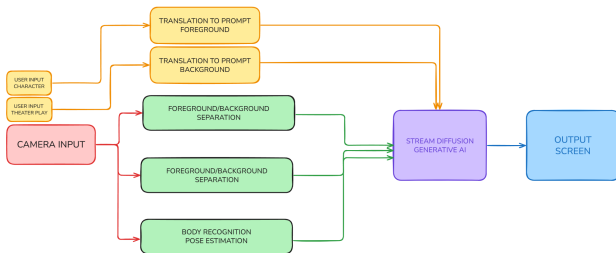


Figure 2: Outline of the system architecture of the interactive generative-AI-based installation *Theater Machine*.

3 Interaction Design and Application Scenarios

The interaction design is intentionally accessible and intuitive, lowering entry barriers and accommodating audiences without prior technical knowledge. Having a low barrier to entry is especially important as this piece is supposed to be publicly available in a theater lobby. Following guidelines in accessible User Interface Design facilitates the use of the installation for a broader range of people, personalities, and capabilities [Guriundefinēdā 2025; Guriundefinēdā and Vatavu 2025].

Approach and Activation: The user enters the detection area in the theater foyer. The system registers their presence and activates the touchscreen interface.

Scene and Character Selection: The user selects from a curated library of canonical theater scenes (e.g., *Swan Lake*, *The Nutcracker*, *Hamlet*) and chooses a character role.

Participatory Immersion: The camera captures the user in real time. The system segments the user's image and integrates it into the AI-generated scene. For example, the user may appear as the "Black Swan" against a digitally rendered *Swan Lake* stage design.

Dynamic Interaction: The system responds to the user's movements in real time, adjusting the visual output dynamically.

Beyond entertainment, the project has a defined educational function. Contextual information about the selected work is displayed on the touchscreen during the interaction. The AI-generated scenes are designed to preserve the artistic essence of the original works, requiring careful model training and parameter control to maintain dramaturgical and aesthetic fidelity [Doğan et al. 2025; Pronin et al. 2024].

4 Outlook and Sustainability Considerations

The scene library is designed for continuous expansion through regular content updates by the *State Theatre of Linz*, increasing long-term utility. The modular hardware and software architecture supports scalability and replication in other theaters or cultural institutions with minimal adaptation, enhancing economic sustainability and potential for broader adoption. The project incorporates a four-dimensional sustainability strategy:

Ecological Sustainability: Selection of energy-efficient hardware and the use of recyclable or reusable materials for housing.

Social Sustainability: Inclusive access through intuitive design, emphasis on accessibility, and gender-neutral content to avoid stereotyping.

Cultural Sustainability: Preservation and reinterpretation of cultural heritage through innovative digital formats, enabling active cultural participation.

Economic Sustainability: Low maintenance requirements after initial installation, and scalability potential.

Gen-AI technology is developing at a rapid pace. In addition to content updates, the system can be made more efficient, and the output rate (frames per second) can increase due to more efficient models and more potent hardware. Preliminary user tests of the prototype have yielded encouraging feedback regarding both usability and artistic engagement. A formal user study is planned as part of future work to empirically evaluate the system's performance and the effectiveness of its design goals.

Acknowledgments

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